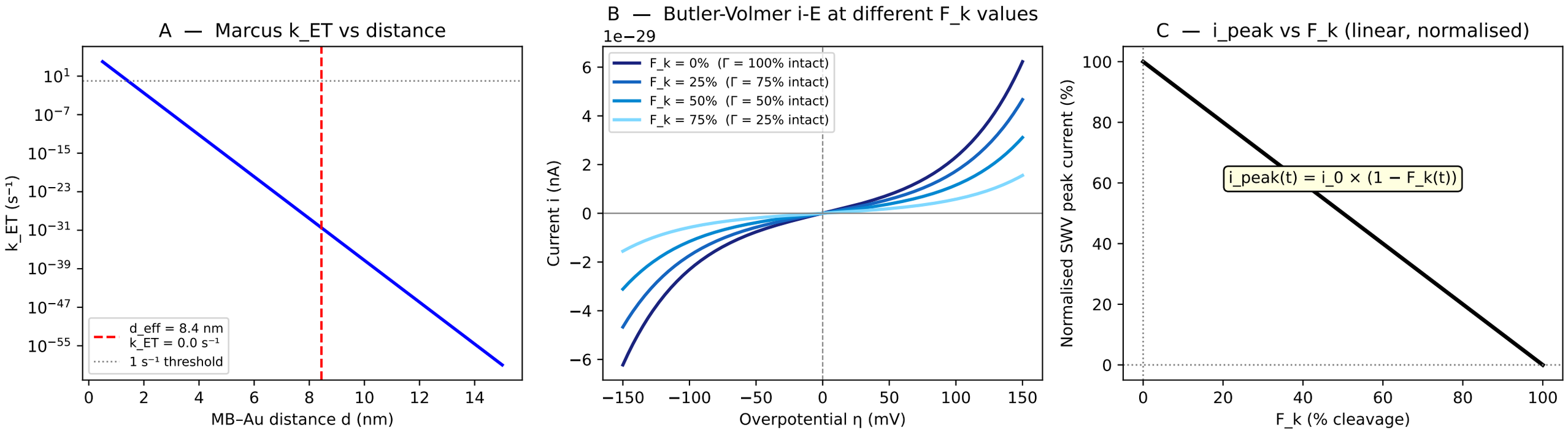
Layer 10: Electron transfer — Marcus model + Butler-Volmer
β = 1.0 Å⁻¹ (CONFIDENT), k_ET_0 = 1e+04 s⁻¹ (FLAG), E°_MB = -250 mV vs Ag/AgCl
A — Marcus k_ET vs distance
k_ET (s⁻¹)
MB–Au distance d (nm)
d_eff = 8.4 nm
k_ET = 0.0 s⁻¹
1 s⁻¹ threshold
B — Butler-Volmer i-E at different F_k values
1e−29
Current i (nA)
Overpotential η (mV)
F_k = 0% (Γ = 100% intact)
F_k = 25% (Γ = 75% intact)
F_k = 50% (Γ = 50% intact)
F_k = 75% (Γ = 25% intact)
C — i_peak vs F_k (linear, normalised)
Normalised SWV peak current (%)
F_k (% cleavage)
i_peak(t) = i_0 × (1 − F_k(t))